

Seminar 7: Frequency Response of FIR Filters

Topics: sinusoidal response of FIR systems, superposition, transient and steady-state behaviour, frequency response, magnitude and phase, periodicity and conjugate symmetry, delay and first-difference filters, cascaded systems, moving-average filters, and filtering sampled continuous-time signals.

Key idea. For an LTI FIR filter, each sinusoidal component keeps its frequency. The frequency response specifies how its amplitude and phase are changed, providing a compact description of the filter in the frequency domain.

1. Sinusoidal response of a simple FIR filter

Consider the FIR system $y[n] = x[n] + 0.5x[n - 1]$

and the complex-exponential input $x[n] = e^{j\hat{\omega}n}$.

- Substitute the input into the difference equation and show that the output can be written as $y[n] = H(e^{j\hat{\omega}})e^{j\hat{\omega}n}$.
- Determine $H(e^{j\hat{\omega}})$.
- Evaluate $H(e^{j\hat{\omega}})$ at $\hat{\omega} = 0, \pi/2$, and π , and give each result in magnitude-phase form.
- For the real input $x[n] = 3\cos(\pi n/2 + \pi/4)$, determine the steady-state output.
- Explain what is meant by saying that a complex exponential is an eigenfunction of an LTI system.

2. From impulse response to frequency response

An FIR filter has impulse response $h[n] = \delta[n] + 2\delta[n - 1] + \delta[n - 2]$.

- Write the corresponding difference equation.
- Derive $H(e^{j\hat{\omega}})$ directly from $h[n]$.
- Factor the frequency response into a real amplitude factor and a linear-phase factor.
- Sketch $|H(e^{j\hat{\omega}})|$ and $\angle H(e^{j\hat{\omega}})$ for $-\pi \leq \hat{\omega} \leq \pi$.
- Identify all frequencies in this interval for which the output is zero.
- State whether the filter behaves qualitatively as a low-pass, high-pass, or neither.

3. Periodicity and symmetry of frequency response

For a real-valued FIR filter with frequency response $H(e^{j\hat{\omega}})$:

- Prove from the definition that $H(e^{j(\hat{\omega}+2\pi)}) = H(e^{j\hat{\omega}})$.
- Show that $H(e^{-j\hat{\omega}}) = H^*(e^{j\hat{\omega}})$.
- Deduce the symmetry properties of the magnitude and phase responses.
- If $|H(e^{j0.4\pi})| = 0.7$ and $\angle H(e^{j0.4\pi}) = -0.3\pi$, determine the magnitude and phase at -0.4π .

- (e) Explain why it is normally sufficient to plot the response over one 2π interval.

4. Pure delay filter

Consider the delay system $y[n] = x[n - n_0]$.

- Determine its impulse response $h[n]$.
- Derive the frequency response $H(e^{j\hat{\omega}})$.
- Determine its magnitude and phase responses.
- Sketch the phase response for $n_0 = 3$ over $-\pi \leq \hat{\omega} \leq \pi$.
- For $x[n] = 2\cos(0.3\pi n + \pi/6)$, write the output for $n_0 = 3$.
- Explain why a pure delay changes phase but not amplitude.

5. First-difference filter

Consider the first-difference system $y[n] = x[n] - x[n - 1]$.

- Write $h[n]$ and derive $H(e^{j\hat{\omega}})$.
- Show that $|H(e^{j\hat{\omega}})| = 2|\sin(\hat{\omega}/2)|$.
- Determine the phase response over $0 < \hat{\omega} < \pi$.
- Find all frequencies at which the response is zero.
- Evaluate the gains at $\hat{\omega} = 0.1\pi$, 0.5π , and π and explain why this filter acts as a simple high-pass filter.
- Determine the response to the unit-step input $u[n]$.

6. Superposition and a multicomponent input

A real FIR filter has frequency-response values $H(e^{j0.1\pi}) = 0.95e^{-j0.2\pi}$, $H(e^{j0.6\pi}) = 0.20e^{-j0.5\pi}$.

The input is $x[n] = 4\cos(0.1\pi n + 0.3\pi) + 2\cos(0.6\pi n - 0.2\pi)$.

- Determine the steady-state output using superposition.
- State the output amplitude and phase of each component.
- Which input component is attenuated more strongly?
- Explain why no new frequencies are created by an LTI FIR filter.

7. Transient and steady-state response

Consider the order-4 FIR filter $h[n] = \{1, -2, 4, -2, 1\}$, $0 \leq n \leq 4$,

with the suddenly applied input $x[n] = \cos(0.2\pi n)u[n]$.

- For which sample indices is the output in the initial transient region?
- Compute $y[0]$, $y[1]$, $y[2]$, $y[3]$, and $y[4]$ directly from the convolution sum.

- (c) Evaluate $H(e^{j0.2\pi})$ and write the steady-state sinusoidal output.
- (d) From what value of n is the steady-state formula valid?
- (e) Explain why the sinusoidal-response rule does not describe the first few output samples exactly.

8. Recovering $h[n]$ and the difference equation from $H(e^{j\hat{\omega}})$

The frequency response of an FIR filter is $H(e^{j\hat{\omega}}) = 2 - 3e^{-j\hat{\omega}} + e^{-j2\hat{\omega}} + 0.5e^{-j3\hat{\omega}}$.

- (a) Determine the impulse response $h[n]$.
- (b) Write the FIR difference equation.
- (c) State the filter order and length.
- (d) Evaluate $H(e^{j\hat{\omega}})$ at DC and at the Nyquist frequency $\hat{\omega} = \pi$.
- (e) Explain the direct correspondence among the coefficient list, $h[n]$, the difference equation, and $H(e^{j\hat{\omega}})$.

9. Cascaded FIR systems in the frequency domain

Two FIR systems are connected in cascade: $h_1[n] = \{1, 1\}$, $h_2[n] = \{1, -1\}$.

- (a) Find $H_1(e^{j\hat{\omega}})$ and $H_2(e^{j\hat{\omega}})$.
- (b) Determine the overall frequency response $H(e^{j\hat{\omega}})$.
- (c) Compute the equivalent impulse response by convolution and verify that it gives the same $H(e^{j\hat{\omega}})$.
- (d) Identify the frequencies at which the cascade has zero response.
- (e) Show that reversing the order of the two systems does not change the overall response.

10. Frequency response of an L-point moving average

An L -point causal moving-average filter is $y[n] = \frac{1}{L} \sum_{k=0}^{L-1} x[n-k]$.

- (a) Derive $H(e^{j\hat{\omega}})$ as a finite geometric series.
- (b) Rewrite it as a product of a real amplitude factor and a linear-phase factor.
- (c) Find the normalized frequencies of all zeros in $0 < \hat{\omega} \leq \pi$.
- (d) What is the gain at DC?
- (e) What delay is represented by the phase factor?
- (f) Explain how increasing L changes smoothing, main-lobe width, the number of zeros, and delay.

11. Comparing even- and odd-length moving averages

Compare a four-point and a five-point causal moving-average filter at normalized frequency $\hat{\omega} = 0.2\pi$.

- (a) Compute the magnitude response of each filter.

- (b) Determine the phase delay, in samples, of each filter.
- (c) Which filter provides stronger smoothing at this frequency?
- (d) Which filter has an integer-sample delay? Which has a fractional-sample delay?
- (e) Explain how a 1.5-sample delay should be interpreted after ideal continuous-time reconstruction.

12. Using a moving average to cancel a sinusoidal disturbance

A measured sequence contains a slowly varying desired signal plus a sinusoidal disturbance at normalized frequency $\hat{\omega} = \pi/2$.

- (a) Find the smallest moving-average length L that has an exact zero at $\hat{\omega} = \pi/2$.
- (b) Write the corresponding difference equation.
- (c) Determine its delay.
- (d) Would the same filter exactly cancel a disturbance at $\hat{\omega} = 0.4\pi$? Explain.
- (e) Suggest a moving-average length that would exactly cancel a component at $\hat{\omega} = 0.4\pi$, if possible.

13. Filtering a sampled continuous-time signal

A continuous-time signal $x(t) = 2\cos(2\pi 50t) + \cos(2\pi 300t)$ is sampled at $f_s = 1000$ samples/s, filtered by a five-point moving average, and ideally reconstructed.

- (a) Determine the normalized radian frequencies of the two sinusoidal components.
- (b) Evaluate the moving-average frequency response at both frequencies.
- (c) Determine the amplitudes and phase shifts of the two reconstructed sinusoidal components.
- (d) Write the reconstructed continuous-time output $y(t)$.
- (e) Explain which analogue component is preserved more strongly and why.
- (f) State the assumption needed for the analogue frequencies to be preserved through sampling and reconstruction.

14. Interpreting a magnitude and phase response

An FIR filter has the following qualitative properties over $0 \leq \hat{\omega} \leq \pi$: unity gain at DC, a smooth decrease in magnitude, a zero at $\hat{\omega} = 2\pi/3$, and approximately linear phase with slope -2 .

- (a) What type of frequency-selective behaviour does this suggest?
- (b) What sinusoidal component is completely removed?
- (c) What time delay is implied by the phase slope?
- (d) Predict qualitatively the output for a signal containing components at 0.1π , $2\pi/3$, and 0.9π .
- (e) Explain why magnitude and phase plots together are needed to predict the complete sinusoidal output.

15. Computational investigation with Python

Use NumPy, Matplotlib, and `scipy.signal.freqz` to investigate the frequency response of FIR filters.

- (a) Plot magnitude and phase for the filters $h_1[n] = \{1, -1\}$, $h_2[n] = \{1/3, 1/3, 1/3\}$, and $h_3[n] = \{1, 2, 1\}$.
- (b) Verify numerically the 2π periodicity and conjugate-symmetry properties for each real FIR filter.
- (c) For each filter, locate any frequency-response zeros and relate them to attenuation of sinusoidal inputs.
- (d) Generate a sum of two sinusoids, filter it with $h_2[n]$, and compare measured output amplitudes with predictions from $|H(e^{j\hat{\omega}})|$.
- (e) Repeat using a suddenly applied sinusoid and identify the transient and steady-state portions of the output.
- (f) Cascade $h_1[n]$ and $h_2[n]$, and verify in both time and frequency domains that convolution of impulse responses corresponds to multiplication of frequency responses.